Final Project – Unit 3: Individual Data Exploration & Dashboard Development

1. Project Overview

This is an **individual project** where each student will select a real-world dataset and perform a complete data analysis workflow. **The project consists of:**

- Exploratory Data Analysis (EDA)
- Data Cleaning & Preprocessing
- Visualizing Insights
- Building an interactive dashboard using Streamlit
- Creating a professional **presentation** summarizing the work
- Submitting a full **PDF report** and **Jupyter Notebook**

This project strengthens your skills in data analysis, visualization, dashboard development, and communication.

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2. Project Deliverables (Required)

Each student must submit:

1. PDF Report

A clean, well-organized report containing all project steps.

2. Jupyter Notebook (.ipynb)

Includes code, outputs, and visualizations.

3. Streamlit App Folder

Contains:

- `app.py`
- Cleaned dataset

- Any helper scripts

4. GitHub Repository Link

Your project must be uploaded to GitHub.

5. Final Presentation (Mandatory)

A 5–7 minute presentation summarizing:

- Dataset
- EDA
- Visualizations
- Dashboard demo
- Key insights

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3. Project Steps (Detailed)

Step 1: Choose a Dataset

You may pick any dataset from:

- Kaggle
- GitHub
- UCI ML Repository
- Any open-source platform

Requirements:

- CSV or JSON
- Any topic (health, education, sales, demographics, etc.)

In your report include:

- Dataset title
- Source link
- Short description (3–4 lines)
- Why you selected it

Step 2: Problem Definition

Write a clear explanation covering:

- What is the dataset about?
- What main questions will you explore?
- What insights do you expect to uncover?

Step 3: Data Loading & Structure

Using Pandas, show:

- First 5 rows (`df.head()`)
- Shape (`df.shape`)
- Data types (`df.info()`)
- Statistics (`df.describe()`)
- Missing values (`df.isnull().sum()`)

Step 4: Data Cleaning

Apply cleaning steps based on your dataset needs:

- Handle missing values
- Remove duplicates
- Convert data types
- Rename columns
- Save cleaned data to CSV

Step 5: Exploratory Data Analysis (EDA)

Perform both **univariate** and **bivariate** analysis.

Univariate

- Histograms
- Countplots
- Boxplots

Bivariate & Multivariate

- Scatter plots
- Bar plots
- Correlation heatmap

Minimum requirement:

At least **5 visualizations** included in the PDF.

Step 6: Insights Summary

Write a short summary describing:

- Patterns you discovered
- Interesting relationships
- Any outliers
- Key takeaways

Step 7: Mini Dashboard (Streamlit)

Build an interactive dashboard using Streamlit.

Sidebar must include:

- Dataset description
- Filters (dropdown, slider, multiselect)

Main page should show:

- Data preview
- Summary statistics
- Interactive visualizations
- Insight sections

Basic structure:

```
import streamlit as st
import pandas as pd
```

```
df = pd.read_csv("clean_data.csv")

st.title("Data Insights Dashboard")

st.sidebar.header("Filters")
```

Optional:

- Add a simple predictive model.

Step 8: Final Summary

In the PDF, include:

- Your main insights
- What you learned
- Limitations of the dataset
- Recommendations for future analysis

Step 9: GitHub Submission

Upload the full project including:

- Notebook
- Cleaned dataset
- Streamlit files
- PDF
- Presentation
- README with instructions

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4. Presentation Requirements (Individual)

Each student will present for **5–7 minutes**.

Your presentation must include:

✔️ **Slide 1 — Title & Name**

✔️ **Slide 2 — Dataset Overview**

✔️ **Slide 3 — Problem Definition**

✔️ **Slide 4 — EDA Results (Plots)**

✔️ **Slide 5 — Dashboard Demo (Screenshots or Live Demo)**

✔️ **Slide 6 — Final Insights**

✔️ **Slide 7 — Conclusion**

5. PDF Required Structure

Your PDF must contain:

1. Title Page
2. Table of Contents
3. Introduction (dataset + problem definition)
4. Data Exploration
5. Data Cleaning
6. EDA (with captions)
7. Dashboard Overview (screenshots)
8. Insights & Conclusion
9. References
